EXPERIMENT 06

Automating Business Processes Using Deluge Scripting and Workflows

Insurance Claim Management

Aim

To automate insurance claim processing in Zoho Creator using Deluge scripts and workflows, and verify the application through positive, negative and repeat-execution tests.

Application and tools

Insurance Claim Management; Claim_Request_Form; Zoho Creator Workflow Builder, Deluge editor, Blueprint designer, reports and function execution logs.

Procedure

1. Use the existing claim form and its related reports. Compute the net payable amount from the repair estimate, depreciation and policy deductible.

2. Configure form submission acknowledgement and validate financial inputs and status changes.

3. Add a reusable missing-document reminder function with a reminder limit and a same-day duplicate guard.

4. Configure high-value review and a Blueprint to control verification, investigation, approval, rejection and settlement.

5. Configure batch reconciliation and recalculate the settlement when the relevant financial values change.

6. Generate a settlement PDF through a reusable function, then test valid records, invalid inputs and repeated execution. Inspect saved fields and audit entries.

Calculation used

Net payable = Repair cost - (Repair cost x Depreciation percent / 100) - Deductible.
Example: INR 38,000 - INR 9,500 - INR 1,000 = INR 27,500.

1. Deluge form workflow

The claim submission workflow assigns the initial processing state and runs the acknowledgement logic. Later controlled tests recorded a new claim reference and an acknowledgement queue marker. A queue marker records that Zoho accepted the send request; inbox receipt was not independently confirmed.

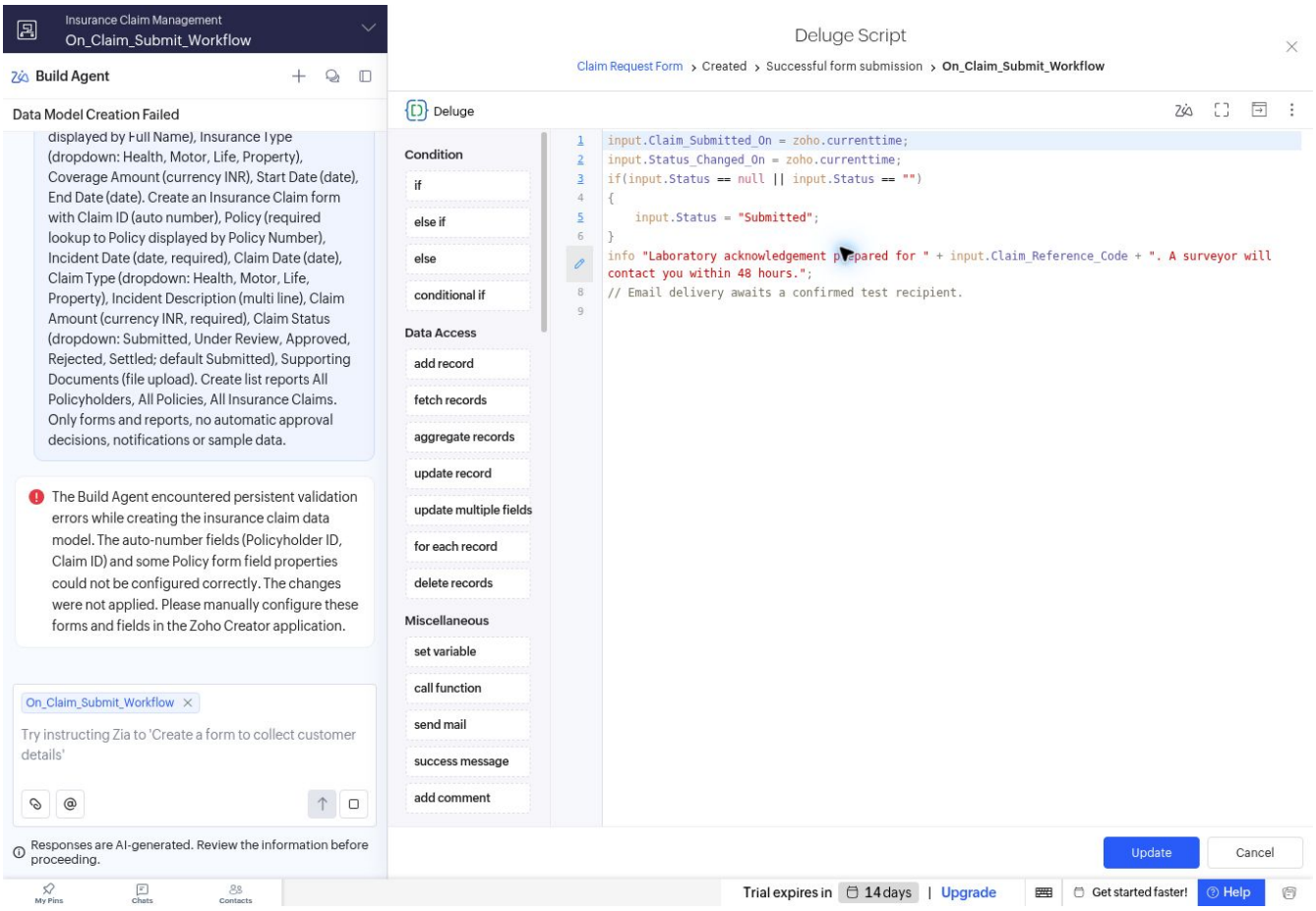


Figure 1. Saved Deluge form workflow configuration captured during implementation. Later runtime checks verify the acknowledgement behaviour.

2. Controlled claim lifecycle

The Blueprint defines Submitted, Document Verification, Investigation, Approved, Settled and Rejected stages. Transition rules require a fresh saved comment. After each successful transition, the script appends an audit entry and clears the comment. Verification transitions allow Write/Administrator; approval, rejection and settlement use Administrator permissions. Separate-account permission testing remains unverified.

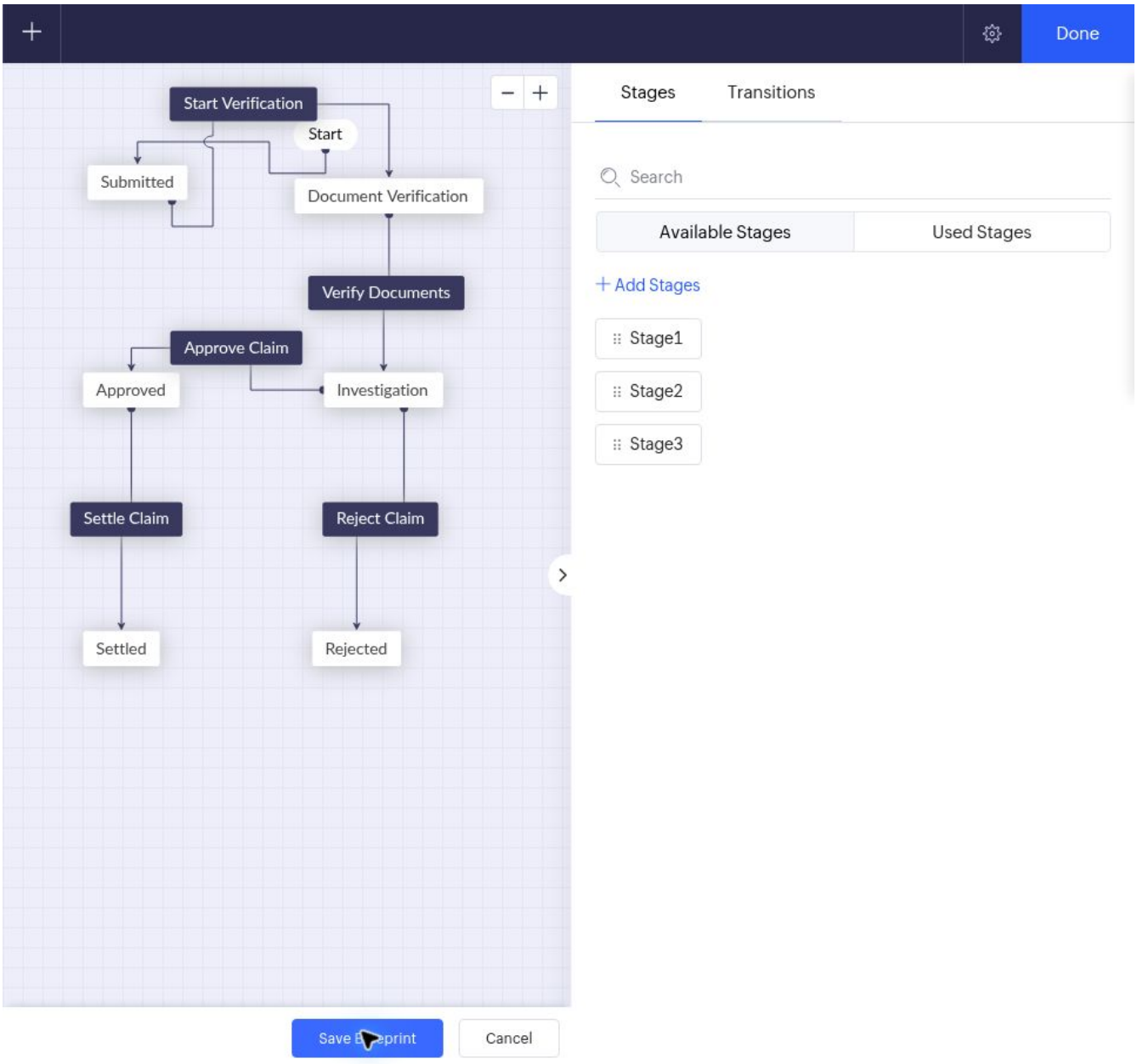


Figure 2. Blueprint stage diagram. The successful lifecycle test reached Settled on the fictional claim ending 000006.

3. Financial validation and high-value review

A repair estimate of INR 70,000, depreciation of 10 percent and a deductible of INR 1,000 produced INR 62,000 and Pending Senior Approval. An empty approval comment was rejected. An owner decision with a fresh comment passed, and the record advanced through approval. Native addition approval uses the existing CEO role; financial edits use the tested owner-only manual review fallback.

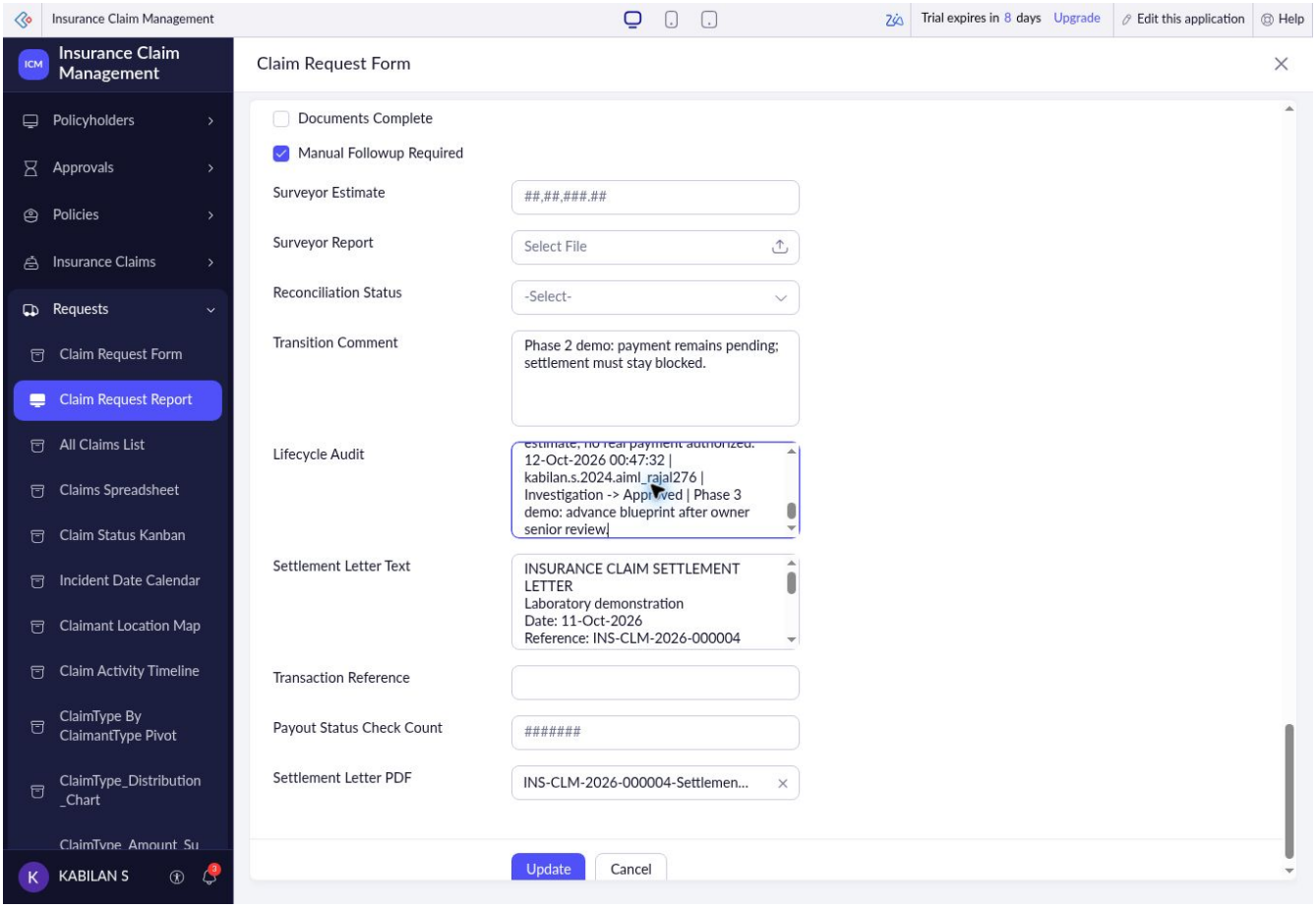


Figure 3. Saved audit evidence from the high-value review test. Exact Senior Claims Manager and independent fallback routing remain unconfigured.

4. Recalculation and reconciliation

Focused tests verified Awaiting Surveyor Report, Matched and Discrepancy outcomes. A fictional surveyor estimate of INR 35,000 against repair cost INR 38,000 produced Discrepancy and manual follow-up. Depreciation 25 percent produced INR 27,500 net; 51 percent was rejected. Single-record batch execution was tested. A complete scheduled nightly run was not observed.

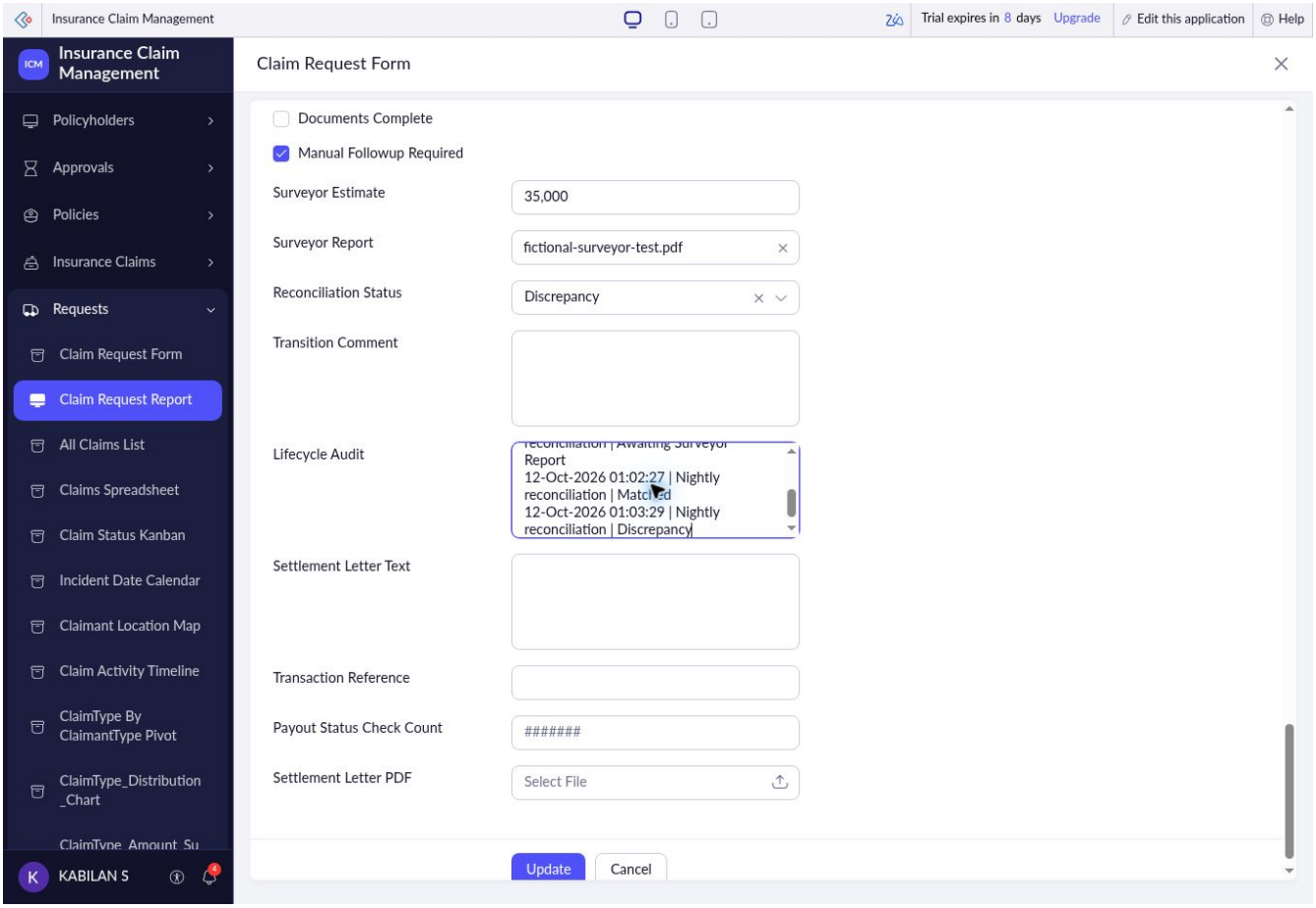


Figure 4. Persisted discrepancy and manual-follow-up fields after a native single-record reconciliation test.

5. Reminder testing and debugging

The reminder function checks claim status, document completion, recipient, reminder count and the date-and-recipient audit marker. One controlled reminder was accepted and the count increased from 0 to 1. Immediate repeat execution returned SKIPPED. Complete-document and five-reminder guard branches were exercised. The daily reminder schedule remains disabled.

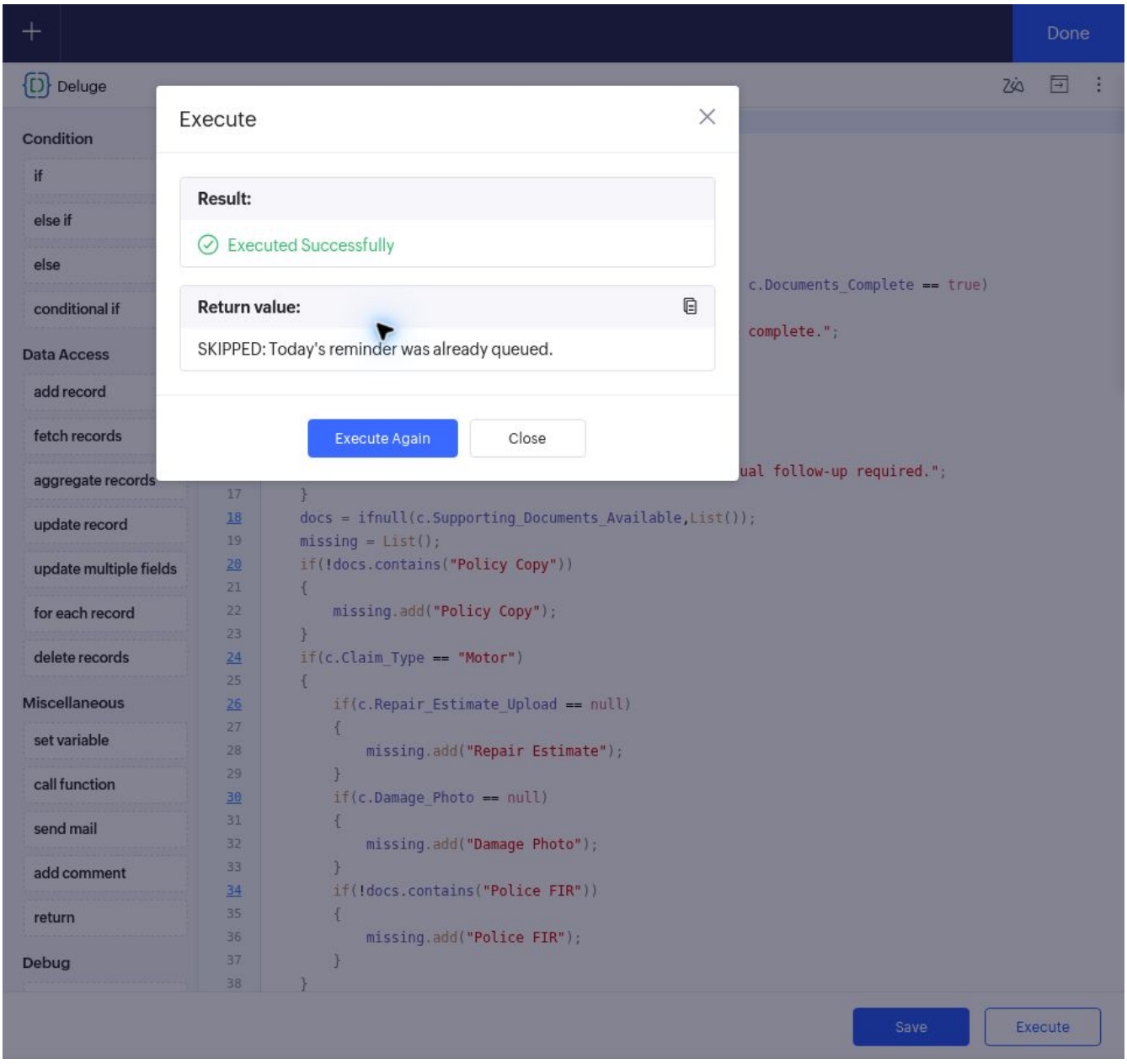


Figure 5. Deluge execution returned SKIPPED because the reminder had already been queued that day. This verifies sequential duplicate prevention.

6. Reusable settlement function

GenerateSettlementLetter delegates to GenerateSettlementLetterWithDelivery. The function validates the record, builds and stores the settlement PDF, and checks Settled plus Paid before sending. The amount-and-recipient audit marker prevents sequential automatic duplicate sends. The function was exercised directly for PDF generation and called through the Blueprint settlement action.

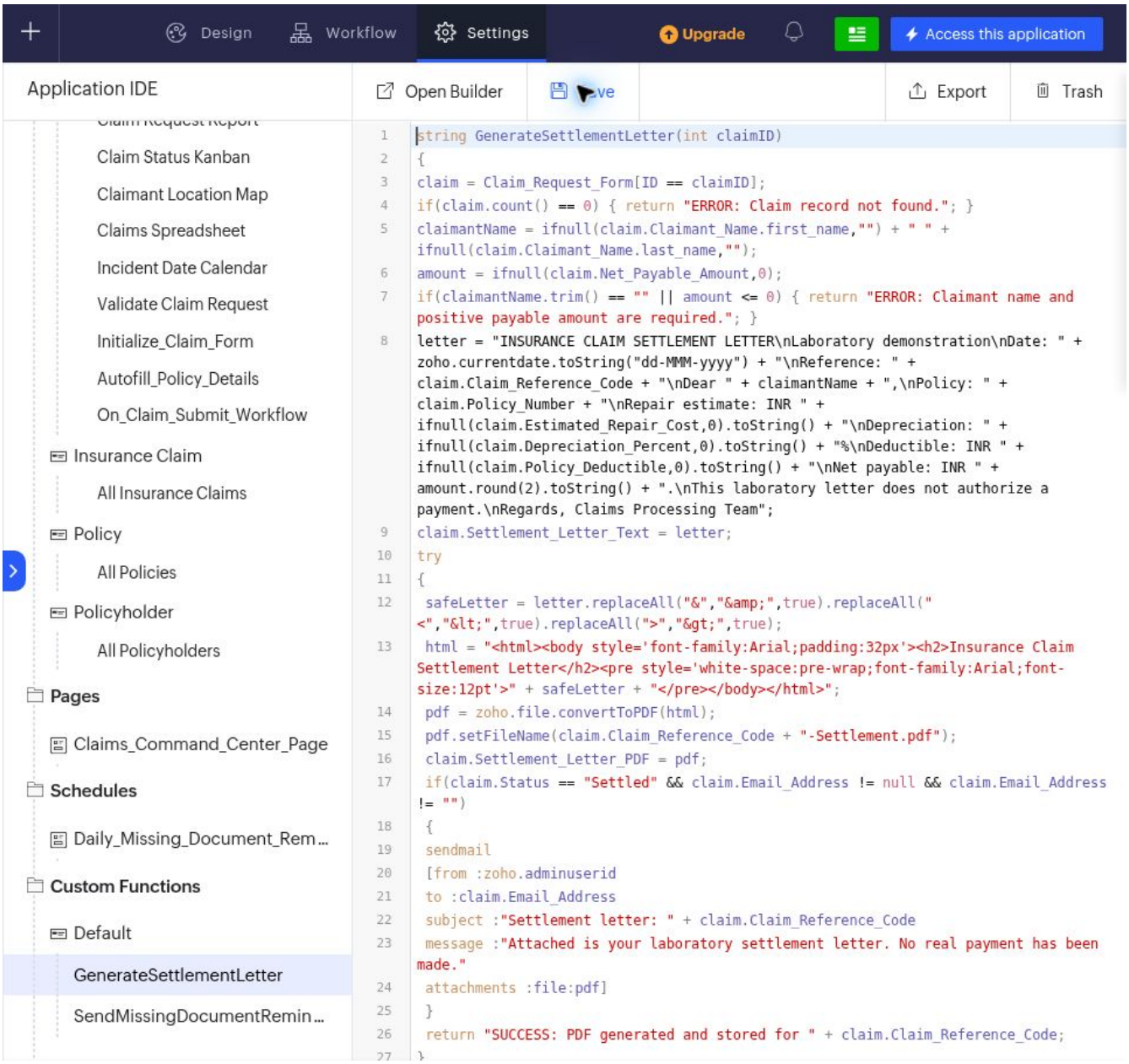


Figure 6. Settlement-letter Deluge function editor captured during implementation. The current delivery helper includes the later send guards.

7. End-to-end settlement evidence

Fictional claim INS-CLM-2026-000006 completed verification, investigation, approval and settlement with net INR 27,500. Its Paid field was explicitly simulated for the laboratory test. The settlement action stored the PDF and returned that the settlement email was queued. No bank transfer occurred. The explicit resend action was saved but its email execution was not completed.

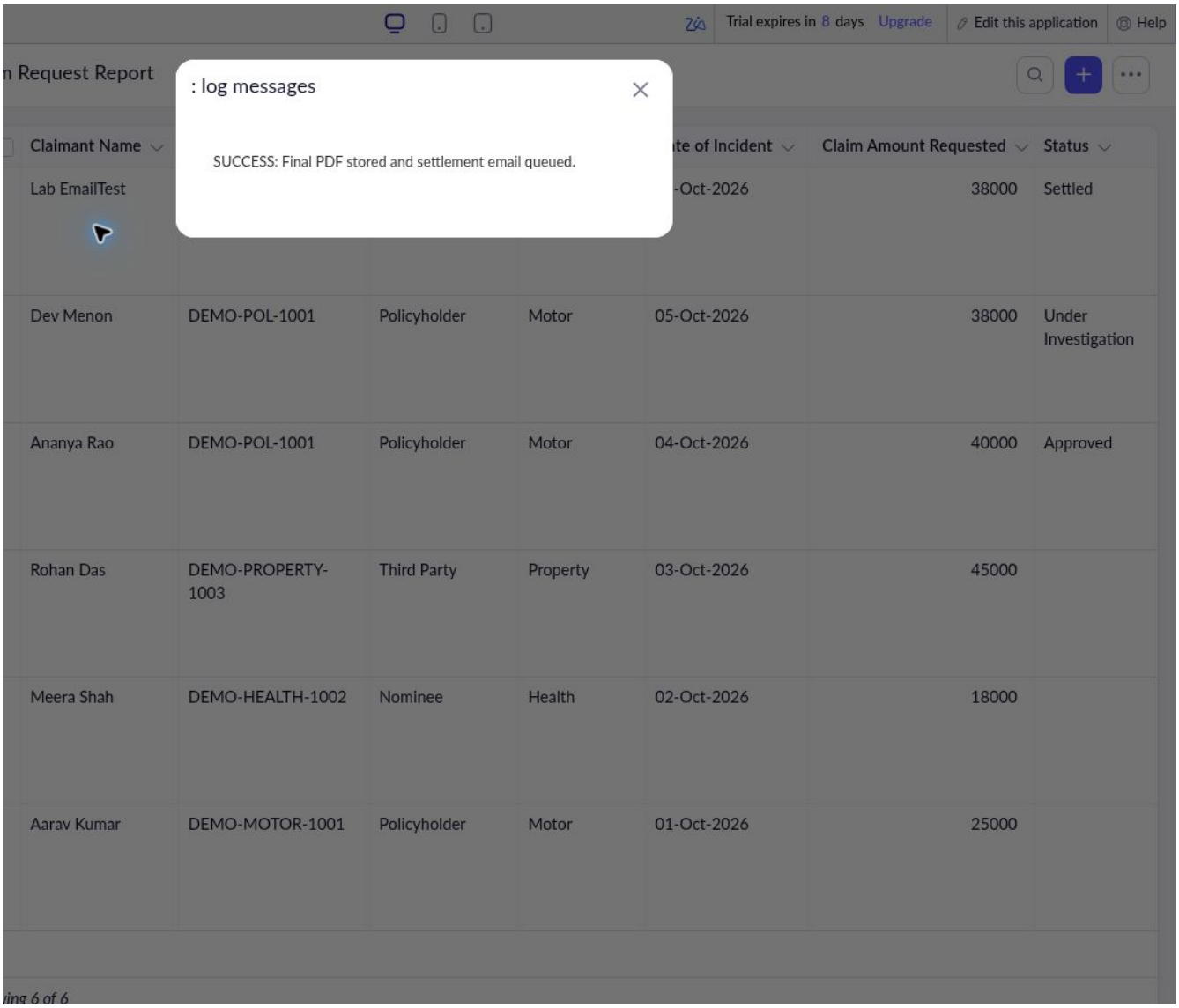


Figure 7. Runtime log confirms the settlement PDF was stored and the email request was queued. Inbox delivery remains unconfirmed.

Test observations

Test	Observed result
Financial recalculation	INR 38,000; 25% depreciation; INR 1,000 deductible => INR 27,500.
Invalid depreciation	51 percent rejected.
High-value edit	INR 62,000 net triggered Pending Senior Approval.
Approval comment	Blank comment rejected; fresh owner decision passed.
Reminder repeat	Same-day second execution skipped; no second reminder queued.
Reminder guard branches	Document-complete and reminder-count-at-five branches exercised.
Reconciliation	Missing report, match and discrepancy cases persisted in focused tests.
Settlement function	PDF generated and stored; invalid record ID rejected.
Lifecycle test	Fictional claim 000006 reached Settled with fresh comments and audit entries.
Email acceptance	Three controlled sends accepted: acknowledgement, reminder and settlement. Inbox receipt unconfirmed.

Result

Business process automation was implemented and tested in the Insurance Claim Management application using Deluge, form workflows, a Blueprint, reusable functions and focused batch execution. Tests demonstrated financial validation, recalculation, high-value review, audit logging, reminder guards and settlement PDF generation.

Scope of verification

This record reuses screenshots and observations from the completed application tests; no new live execution was performed while preparing it. Real payout integration remains paused. Production payout callbacks, finance notifications, independent approver routing, a full nightly batch run and concurrent execution are outside the verified results above.